

SELQIE

Standard Operating Procedure

V1.0

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for use with selqie_ws version 3.0

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Safety Notes

2 people must always be present for the powering, operation, and testing of SELQIE.

Batteries

The batteries used to run SELQIE's motors are labeled #2 and #3. They are connected in parallel while powering the motor system. However, they should be removed from the parallel connection harness during storage.

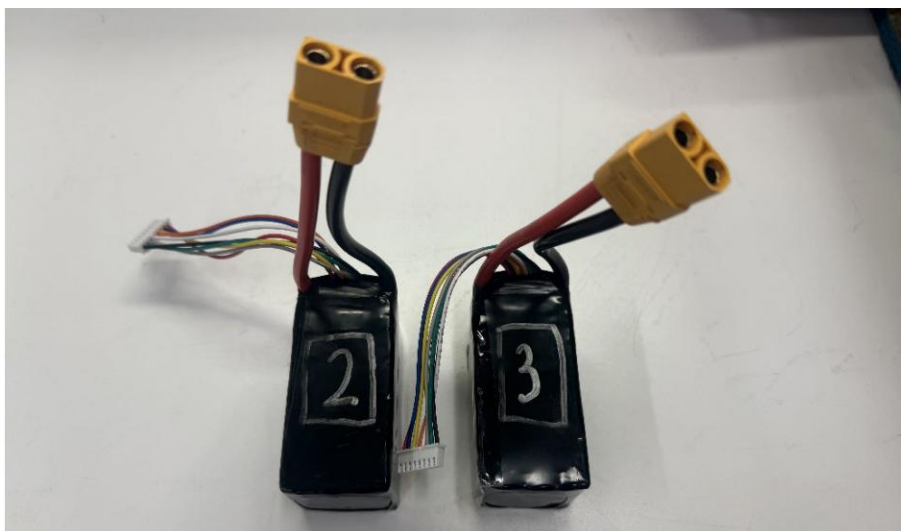


Figure 1: SELQIE motor batteries (#2&3)

E-Stop

The motor power connection to SELQIE features an emergency stop (E-Stop) button. This button should be pressed in the event of concerning electrical or motor behavior with SELQIE. This includes but is not limited to: erratic leg movement, presence of smoke, motors running uncontrollably, and someone/something present within the legs' paths of motion.



Figure 2: Motor Power Emergency Stop

Wiring

Wire connections should always be inspected **prior to** connecting SELQIE to power. Check for loose connections, dismantled electrical components, or unintended contact of bare metal. The 3.3V/Ground Bus (Figure 3) should have all proper female headers fully secure in the housing. If you are not **100%** confident that the wiring looks correct, ask another person to check the wiring behind you. If any suspicion continues, **do not power SELQIE**.

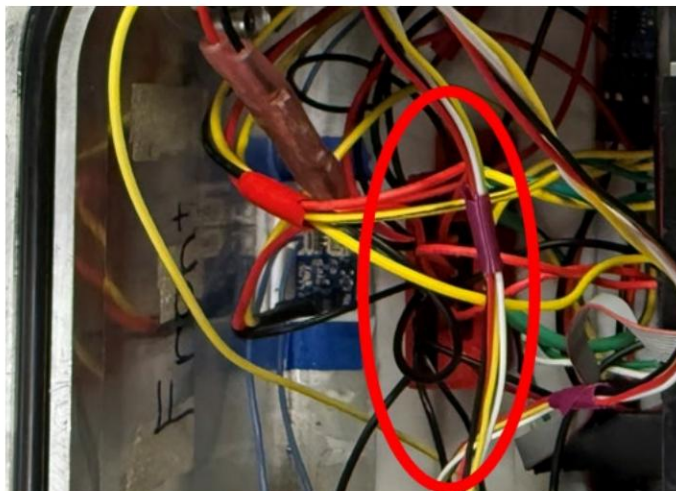


Figure 3: Logic 3.3V/Ground Bus Location

Water

Due to the nature of aquatic robotics, SELQIE should be fully inspected to ensure that water is not present inside. This includes visually checking the main body and listening to the hip enclosures for the presence of water. If you have any suspicion, get a second opinion.

Leak sensors are present within each hip, and within the main body. In the event of water presence, the leak sensors will trigger the Orin to shut down. If the Orin shuts down at any time unexpectedly, power should be **immediately** disconnected and, if applicable, SELQIE should be retrieved from the water.

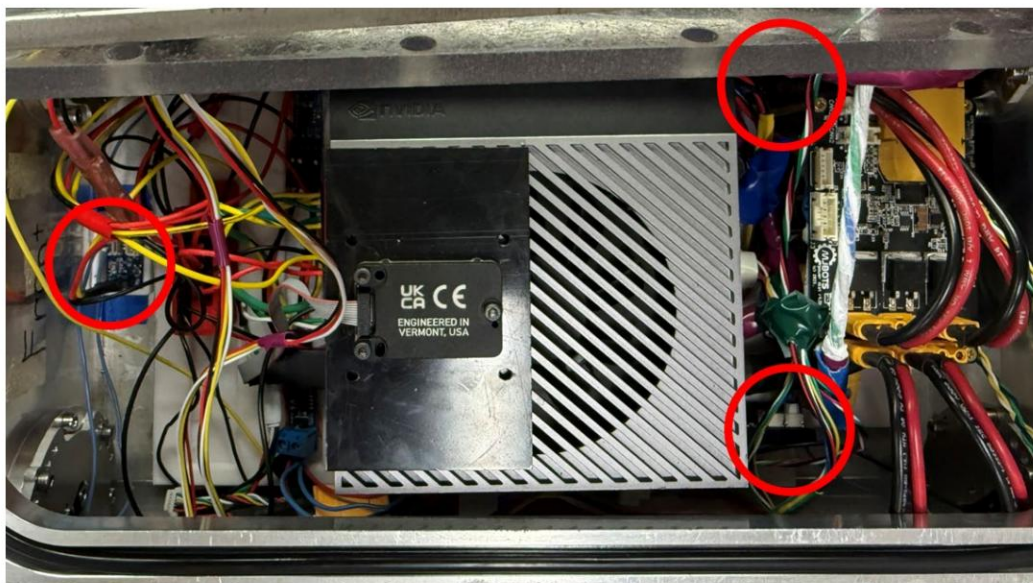


Figure 4: Leak Sensor Board Locations

Testing

Apply Zero Guides

SELQIE should be powered on in a specific order. Before powering, ensure that the green zeroing blocks are attached to each leg. They should clip on comfortably over each leg, attaching to the top and bottom of each hip without much force. The side of each guide with the deeper groove will go over the outermost side of the leg's linkage.

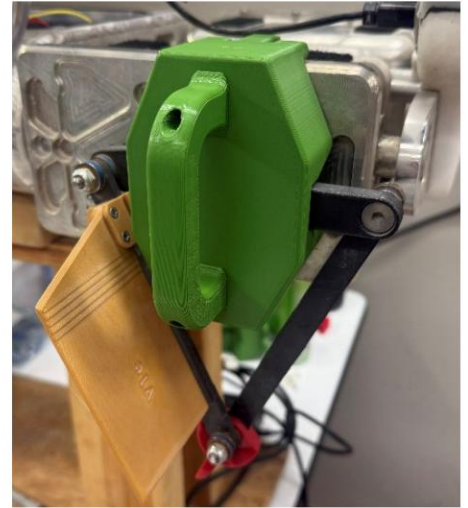


Figure 5: Applied Zeroing Guide on Front Right Leg

Powering on SELQIE

Motor Power

Once the zero guides are applied, batteries #2&3 can be connected into the parallel connection harness (Figure 4). Once the batteries are in the harness, ensure that the E-Stop is not activated. Then the remaining harness terminal can be connected to the E-Stop (Figure 5).

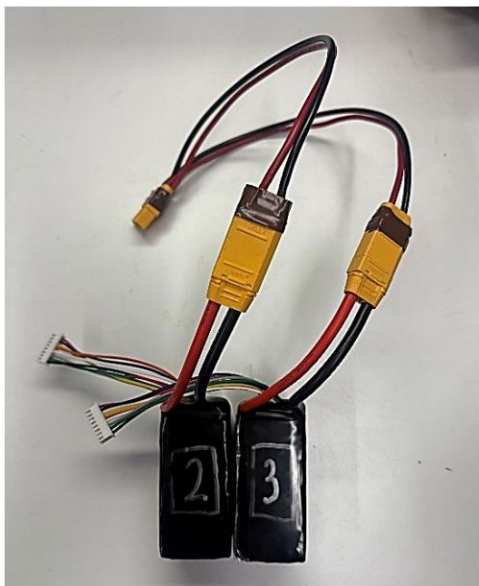


Figure 6: Batteries #2&3 in Motor Power Harness

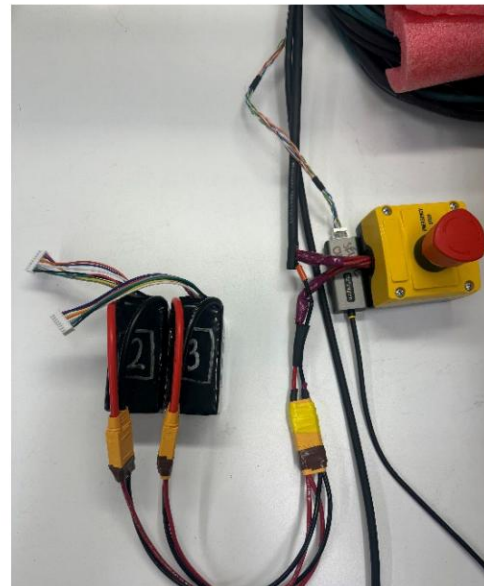


Figure 7: Motor Power Harness Fully Connected

A red light should be present on the lower power distribution board. It should be inspected for any abnormal behavior.

Logic Power

SELQIE's logic circuit is powered using the designated wall power adapter located at the ONR Project Desk. To begin, the wall side of the power cable should be disconnected at the adapter block. While keeping the middle disconnected, the 2 outer ends of the power cord can be plugged into the SELQIE and the wall, respectfully (Figure 8).

Then, the middle portion of the adapter can be connected. This is done to minimize spark risk. A red light should now be present on the upper power distribution board, and it should be inspected for abnormal behavior. Wait 60 seconds before connecting to SELQIE from the laptop.



Figure 8: Logic Power Cord

Connecting to SELQIE

To connect to SELQIE's onboard computer, the ethernet cable on should be connected to the laptop via the USB-C to ethernet adapter. Ensure that unnecessary stress is not put onto the tether. In some cases, this may require taping the non-SELQIE side of the tether down to your working surface.

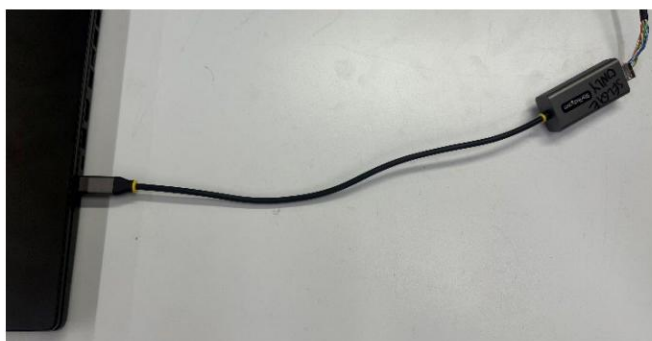


Figure 9: USB-C to Ethernet Adapter connected to the SELQIE laptop and tether

To connect to SELQIE from the SELQIE laptop, type the following code into terminal:

```
>ssh selqie
```

When prompted for a password, enter the following:

```
> [REDACTED]
```

At this point, you should now be inside SELQIE's home directory.

Launching *SELQIE TERMINAL*

To launch selqie terminal, first navigate to the tmux folder of the workspace.

```
>cd selqie_ws/src/tmux
```

Run the selqie.sh file using the command below.

```
>./selqie.sh
```

About the Test Environment

The test environment consists of four windows.

1. Top left window: status updates.
2. Top right: SELQIE TERMINAL- Used for controlling the test.
3. Bottom left: standard Linux terminal.
4. Bottom right: Jetson GUI

Running a test

Use SELQIE TERMINAL (top right)

Before removing the zero guides (Figure 5), run the zero command.

```
>zero
```

Remove the zero guides. Once removed and legs are clear of wires or obstacles, run the ready command.

```
>ready
```

For data recording, the “rosbags” data processor can be initiated to save all relevant topic data. Recording data will be saved with the date/timestamp that recording was started. They will be saved to */home/rosbags/*.

```
>start_recording
```

The legs should now be in the stance position. To run feed forward control, use the run trajectory function.

```
>run_trajectory <gait file> <# of loops> <hz>
```

The gait file can be: walk.txt, jump.txt, crawl.txt, or swim.txt.

Data recording can be stopped using the command below.

```
>stop_recording
```

To exit the control interface, type exit.

>exit

Shutting Down the Orin

In a terminal window, run the following code to shut down the Orin.

>sudo shutdown -h now

Enter the password

> XXXXXXXXXX

Powering off SELQIE

SELQIE should be powered off in a specific order. The motor battery harness is disconnected from the SELQIE tether first. The lower power distribution board should be inspected for any abnormal behavior; the red light should be off.

The individual batteries should then be removed from the motor power harness. After confirming that motor power is disconnected, the logic power source (wall power) can be disconnected from the middle block and should no longer display a red light on its power distribution board. The other ends of the cord can then be disconnected from SELQIE and the wall.